Problem Set 4

Applied Stats II

Due: April 10, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Friday April 10, 2026. No late assignments will be accepted.

Question 1

We're interested in modeling the historical causes of child mortality. We have data from 26855 children born in Skellefteå, Sweden from 1850 to 1884. Using the "child" dataset in the `eha` library, fit a Cox Proportional Hazard model using mother's age and infant's gender as covariates. Present and interpret the output.

Question 2

We want to estimate how the amount of damage caused by a disaster relates to (1) whether disaster relief is provided and (2) the amount of relief that is provided conditional on a donation occurring when a disaster hits a given country. Estimate a Heckman selection model using the `disasters_response` data on GitHub in which `binContribution` is the outcome for the selection equation, and `originalContributionMillionUSDLogged` is the outcome for the second equation. Use `occurrences + deathsEM + normalizedDamageEMLogged` as your input variables for both equations. Present and interpret the output.

My Answers

Question 1

To fit the Cox proportional hazard model, I first kept only the variables needed, then ran the model:

```r
# fit Cox proportional hazards model
child_cox_model <- coxph(Surv(enter, exit, event) ~ m.age + sex, data = child_data)

# model output
summary(child_cox_model)
```

The model provides me with the following output:

Table 1: Cox Proportional Hazards Model of Child Mortality

	Dependent variable:
	Hazard of child death
Mother's age	0.008***
	(0.002)
Female	−0.082***
	(0.027)
Observations	26,574
R^2	0.001
Max. Possible R^2	0.986
Log Likelihood	−56,503.480
Wald Test	22.520*** (df = 2)
LR Test	22.518*** (df = 2)
Score (Logrank) Test	22.530*** (df = 2)
Note:	*p<0.1; **p<0.05; ***p<0.01

The model output shows that the model is overall statistically significant. The LR test with 2 degrees of freedown equals 22.5 ($p < 0.01$), thus indicating that for this dataset, both the mother's age and the gendere of the infant are associated with the hazard of child mortality.

Mother's age has a positive and statistically significant effect ($\hat{\beta} = 0.008$, $p < 0.01$). On the hazard ratio scale, this corresponds to $\exp(0.008) = 1.008$. This means that each additional year of the mother's age is associated with about a 0.8% increase in the hazard of child death, holding infant gender constant. The 95% confidence interval for the hazard ratio is approximately $[1.003, 1.012]$, which excludes 1.

Infant gender is also statistically significant. Using male infants as the reference group, the coefficient for female infants is negative ($\hat{\beta} = -0.0822$, $p < 0.01$). The corresponding hazard ratio is $\exp(-0.0822) = 0.921$, indicating that female infants have a lower hazard of death than male infants. Specifically, the hazard for female infants is about 7.9% lower than for male infants, holding mother's age constant. The 95% confidence interval $[0.874, 0.971]$ also excludes 1.

Overall, the results suggest that children born to older mothers face a slightly higher mortality risk, while female infants face a lower risk compared to male infants. Although statistically significant, the effect of mother's age is relatively small in substantive terms on a per-year basis.

Question 2

I estimated a Heckman selection model to account for the fact that contributions are only observed when a donation occurs. The selection equation models whether any disaster relief is provided, while the outcome equation models the logged amount of relief conditional on a donation occurring. The model provided me with the following output:

Table 2: Heckman Selection Model (Tobit Type II)

Selection Equation (Probit)				
Variable	Estimate	Std. Error	t value	$\Pr(>\lvert t\rvert)$
Intercept	-1.3039	0.0180	-72.34	$< 2 \times 10^{-16}$
occurrences	0.0193	0.0017	11.65	$< 2 \times 10^{-16}$
deathsEM	0.0119	0.0007	16.23	$< 2 \times 10^{-16}$
normalizedDamageEMLogged	0.0210	0.0009	23.29	$< 2 \times 10^{-16}$
Outcome Equation				
Variable	Estimate	Std. Error	t value	$\Pr(>\lvert t\rvert)$
Intercept	0.4135	0.4075	1.015	0.3102
occurrences	-0.0018	0.0064	-0.291	0.7710
deathsEM	0.0058	0.0020	2.856	0.0043
normalizedDamageEMLogged	-0.0181	0.0052	-3.450	0.0006
Error Terms				
Parameter	Estimate	Std. Error	t value	$\Pr(>\lvert t\rvert)$
σ	1.9360	0.1038	18.65	$< 2 \times 10^{-16}$
ρ	-0.5267	0.0910	-5.79	7.29×10^{-9}

Log-Likelihood: -17557.78
Observations: 49,096 (45,848 censored, 3,248 observed)
Free parameters: 10

Looking first at the *selection equation*, all three explanatory variables have positive coefficients and are highly statistically significant. The estimates for `occurrences` (0.0193), `deathsEM` (0.0119), and `normalizedDamageEMLogged` (0.0210) all have $p < 0.001$. This suggests that disasters that happen more frequently, cause more deaths, and result in higher economic damage are more likely to receive some form of aid.

Turning to the *outcome equation*, the findings are less consistent. The coefficient on `occurrences` is close to zero and not statistically significant ($\hat{\beta} = -0.0018$, $p = 0.771$), which implies that the number of occurrences does not seem to influence the size of the donation once aid is given. In contrast, `deathsEM` is positive and significant ($\hat{\beta} = 0.0058$, $p = 0.004$), indicating that disasters with higher mortality tend to receive larger contributions, conditional on receiving aid. Interestingly, `normalizedDamageEMLogged` has a negative and significant coefficient ($\hat{\beta} = -0.0181$, $p < 0.001$), suggesting that, among disasters that receive aid, higher levels of logged economic damage are associated with slightly smaller contribution amounts.

The estimate of the correlation between the error terms is negative and statistically significant ($\hat{\rho} = -0.527$, $p < 0.001$). This indicates that unobserved factors influencing the likelihood of receiving aid are related to unobserved factors affecting the amount of aid. As a result, using a Heckman selection model is justified instead of relying on a standard regression based only on observed donations. The estimated standard deviation of the outcome equation error term is $\hat{\sigma} = 1.936$.

Overall, the results show that more severe disasters are more likely to receive assistance, particularly those with higher death tolls and greater economic impact. However, once aid is provided, the size of the contribution appears to depend more on mortality than on how often disasters occur, while higher economic damage is linked to a small decrease in the logged amount of aid.

Code used:

```
disaster <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsII_2026/refs/heads/main/datasets/disaster_response.csv")

# retain only the variables required for Question 2
disaster_clean <- disaster[, c(
  "binContribution",
  "originalContributionMillionUSDLogged",
  "occurrences",
  "deathsEM",
  "normalizedDamageEMLogged"
)]

# inspect the main distributional features and check for missing values
summary(disaster_clean)
colSums(is.na(disaster_clean))

# verify that binContribution is coded as 0/1 and inspect the class balance
```

```
table(disaster_clean$binContribution)
# -> the classes are uneven, but not enough to cause major concern here

# check whether the logged contribution variable uses -25.328 as a marker for no donation
tapply(
  disaster_clean$originalContributionMillionUSDLogged,
  disaster_clean$binContribution,
  summary
)

# -> when binContribution = 0, the logged contribution always equals -25.328
# -> when binContribution = 1, the values show real variation

# in the Heckman setup, the outcome is only defined for observations with selection = 1
# therefore, assign NA to contribution amounts when no donation was made
disaster_clean$originalContributionMillionUSDLogged[
  disaster_clean$binContribution == 0
] <- NA

# keep observations with valid selection and predictor data;
# missing outcome values are acceptable only when selection = 0
disaster_clean <- subset(
  disaster_clean,
  !is.na(binContribution) &
    !is.na(occurrences) &
    !is.na(deathsEM) &
    !is.na(normalizedDamageEMLogged) &
    !(binContribution == 1 & is.na(originalContributionMillionUSDLogged))
)

# recode the selection indicator as a binary factor
# the value "1" represents cases where a donation is observed
disaster_clean$binContribution <- factor(disaster_clean$binContribution, levels = c(0, 1))

# estimate the Heckman selection model using maximum likelihood
disaster_heckman <- selection(
  selection = binContribution ~ occurrences + deathsEM + normalizedDamageEMLogged,
  outcome   = originalContributionMillionUSDLogged ~ occurrences + deathsEM + normalizedDamageEMLogged,
  data      = disaster_clean,
  method    = "ml"
)

# display the complete set of model results
summary(disaster_heckman)
```